TITANIC DATASET

END-TO-END DATA ANALYTICS CASE STUDY

Internship Final Project – Task 4

Submitted By: Alsaba Farooque Bhairamadgi

Organization: Swynex Technologies

Tools Used:

- Python
- Pandas
- Microsoft Excel
- Microsoft Power BI

Project Domain: Data Analytics

Academic Year: 2026–2027

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1. INTRODUCTION

Data Analytics is the process of collecting, cleaning, analyzing, and visualizing data to find useful information.

This project focuses on the Titanic dataset, which contains information about passengers who travelled on the Titanic ship.

The main purpose of this project is to analyze passenger details and understand the factors that influenced survival.

The project uses Python, Pandas, Microsoft Excel, and Power BI for data cleaning, analysis, and visualization.

2. PROBLEM STATEMENT

The Titanic dataset contains passenger information such as age, gender, passenger class, fare, and survival status.

The main problem is to identify which factors influenced the survival of passengers during the Titanic disaster.

This project analyzes passenger data to find survival patterns and understand the relationship between passenger characteristics and survival rates.

3. PROJECT OBJECTIVES

The main objectives of this project are:

1. To understand the Titanic passenger dataset.
2. To identify and handle missing values.
3. To remove duplicate records.
4. To perform Exploratory Data Analysis (EDA).
5. To analyze passenger survival patterns.
6. To create charts and an interactive Power BI dashboard.
7. To identify meaningful business insights from the data.

4. DATASET INFORMATION

The Titanic dataset is used for this project. It contains passenger details from the Titanic ship, including age, gender, passenger class, fare, and survival status.

Dataset Details:

- Dataset Name: Titanic Dataset
- Original Rows: 891
- Total Columns: 15
- Final Rows After Cleaning: 781
- Data Format: CSV
- Data Type: Structured Data

Main Attributes:

- Passenger ID
- Survived
- Pclass
- Sex
- Age
- Fare
- Embarked

5. DATA CLEANING PROCESS

Data cleaning was performed using Python and Pandas in Google Colab to improve the quality of the Titanic dataset.

Cleaning Steps:

1. Identified missing values in the dataset.
2. Filled missing Age values using the median.
3. Filled missing Embarked values using the mode.
4. Replaced missing Deck values with "Unknown".
5. Removed duplicate records.
6. Standardized text values and data types.
7. Validated the cleaned dataset.

Final Cleaning Results:

Description	Result
Original Rows	891
Final Rows	781
Total Columns	15
Missing Values	0
Duplicate Records	0

```
[ ]
print("FINAL DATA CLEANING REPORT")
print("--------------------------")

print("Rows:", cleaned_df.shape[0])
print("Columns:", cleaned_df.shape[1])
print("Total missing values:", cleaned_df.isnull().sum().sum())
print("Duplicate records:", cleaned_df.duplicated().sum())

print("\nMissing values by column:")
print(cleaned_df.isnull().sum())

print("\nFinal validation completed!")
```

```
FINAL DATA CLEANING REPORT
--------------------------
Rows: 781
Columns: 15
Total missing values: 0
Duplicate records: 0

Missing values by column:
survived       0
```

6. EXPLORATORY DATA ANALYSIS (EDA)

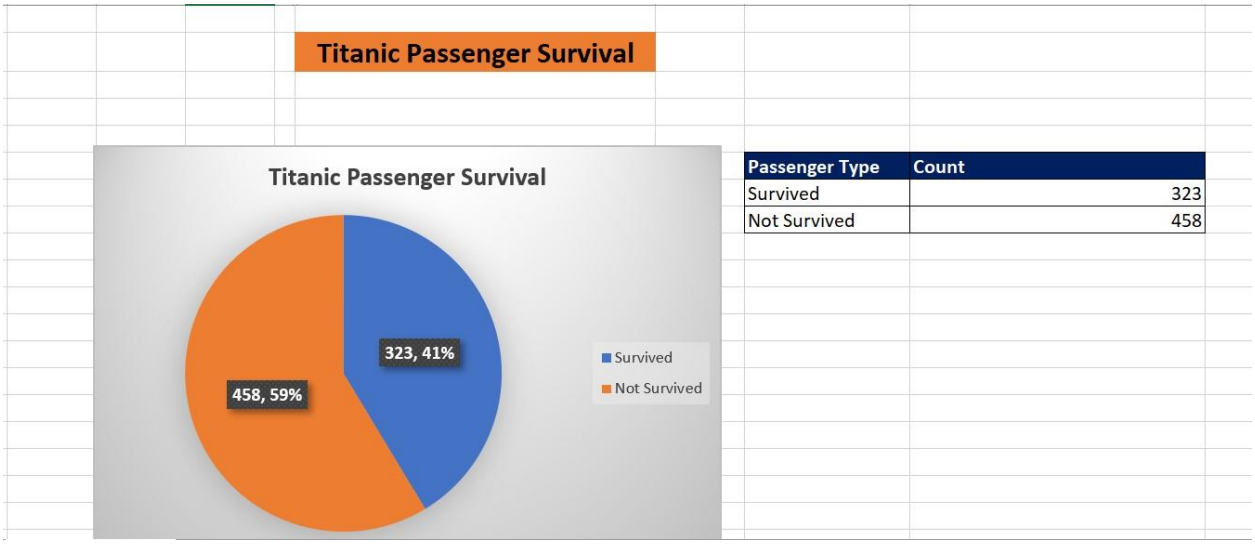
Exploratory Data Analysis was performed using Microsoft Excel to understand passenger information and identify survival patterns.

Analysis Performed:

1. Analyzed passenger distribution by gender.
2. Compared survival and non-survival counts.
3. Studied passenger class distribution.
4. Examined age-wise passenger information.
5. Analyzed survival patterns using charts.

Purpose of EDA:

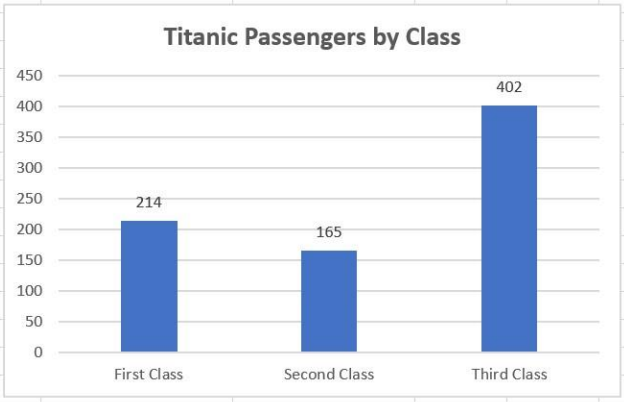
The purpose of EDA is to identify patterns, trends, and relationships in the Titanic dataset through data analysis and visualizations.



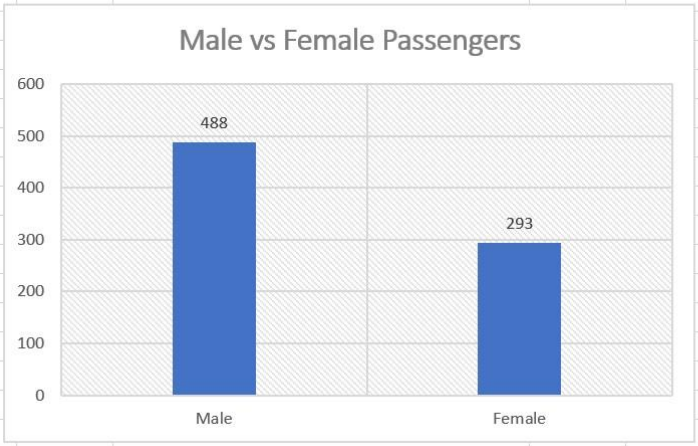
Passenger Type	Count
Survived	323
Not Survived	458

Titanic Passengers by Class

Passenger Class	Count
First Class	214
Second Class	165
Third Class	402

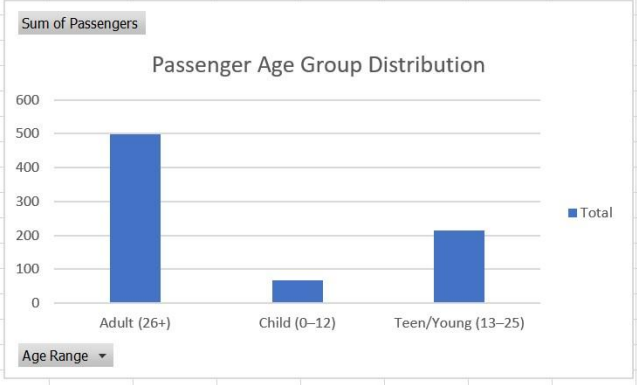


Male vs Female Passengers



Gender	Count
Male	488
Female	293

Row Labels	Sum of Passengers
Adult (26+)	498
Child (0–12)	68
Teen/Young (13–25)	215
Grand Total	**781**



7. DATA VISUALIZATION

Data visualization was performed using Microsoft Excel and Power BI to present the analysis in a clear and understandable format.

The following visualizations were created:

1. Passenger distribution by gender.
2. Survival and non-survival comparison.
3. Passenger class distribution.
4. Age group analysis.
5. Survival analysis using charts.

These visualizations help identify important patterns and make the analysis easier to understand.

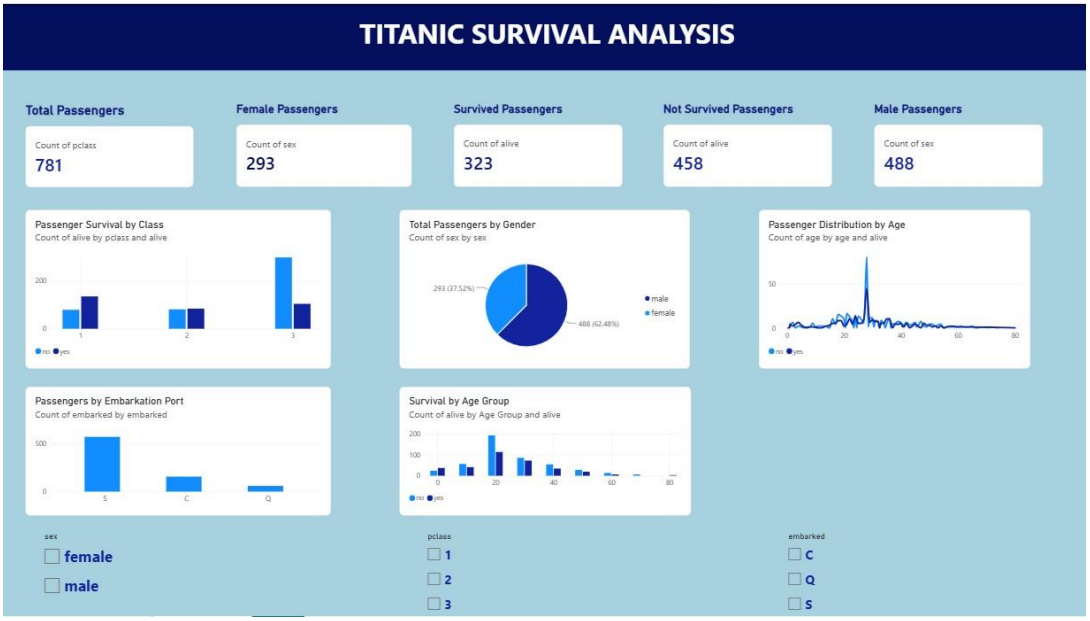
8.POWER BI DASHBOARD

An interactive dashboard was created using Microsoft Power BI to analyze Titanic passenger survival.

Dashboard Components:

1. Total Passengers KPI Card
2. Female Passengers KPI Card
3. Survived Passengers KPI Card
4. Not Survived Passengers KPI Card
5. Male Passengers KPI Card
6. Passenger survival charts
7. Interactive slicers for filtering data

The dashboard helps users explore passenger information and understand survival patterns through interactive visualizations.



9.KEY BUSINESS INSIGHTS

The analysis of the Titanic dataset revealed the following insights:

1. Total passengers analyzed: 781.
2. Total survived passengers: 323.
3. Total non-survived passengers: 458.
4. Female passengers: 293.
5. Male passengers: 488.
6. The survival analysis shows differences based on passenger gender and class.

These insights help understand passenger survival patterns and the factors associated with survival.

10. CONCLUSION

This project successfully analyzed the Titanic dataset using Python, Pandas, Microsoft Excel, and Power BI.

The data cleaning process improved data quality by handling missing values and removing duplicate records.

Exploratory Data Analysis and visualizations helped identify passenger survival patterns.

The interactive Power BI dashboard presented the results in a clear and understandable way.

Overall, this project improved practical knowledge of data cleaning, data analysis, visualization, and business insights.

11. TOOLS AND TECHNOLOGIES

The following tools and technologies were used to complete this project:

1. **Python:** Used for data cleaning and preprocessing.
2. **Pandas:** Used for handling and analyzing data.
3. **Google Colab:** Used to execute Python code.
4. **Microsoft Excel:** Used for Exploratory Data Analysis and charts.
5. **Microsoft Power BI:** Used to create an interactive dashboard.
6. **GitHub:** Used to store and share project files.

12. GITHUB REPOSITORY

The complete project files are available on GitHub.

GitHub Repository:

https://github.com/alsababahiramadgi2007-maker/titanic-survival-analysis-dashboard

The repository contains the Power BI dashboard and project documentation.

The data cleaning notebook and cleaned dataset are also available in the Task 1 repository.